

Predicting translation BLEU from monolingual competence: an additive decomposition

1 Introduction

A central question for multilingual large language models is the extent to which translation quality is determined by per-language competence alone, as opposed to language-pair-specific factors. We frame this as a specific hypothesis:

H1. Translation BLEU between an ordered language pair (i, j) is predictable from per-language monolingual competence on i and j , without a substantive language-pair-specific interaction term.

We address H1 via a per-language decomposition of pairwise BLEU scores into source and target effects, and we test whether those effects are predicted by monolingual competence proxies and whether any pair-specific residual is detectable.

Our results support H1:

1. An additive random-effects model $\text{BLEU}_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$ explains 86–93% of centered BLEU variance across two models. A multiplicative (rank-1) decomposition fits comparably, but does not outperform additive on the same parameter budget.
2. The recovered language effects α, β correlate with standard monolingual competence proxies (FLORES corpus perplexity and Multi-BLiMP grammatical-acceptability accuracy) in the expected direction.
3. Monolingual competence proxies directly predict BLEU at the pair level. Multi-BLiMP accuracy explains up to 44% of BLEU variance on its own; FLORES PPL is a substantially weaker proxy.
4. Source (α) and target (β) effects contribute asymmetrically to BLEU. Target-language competence dominates for Llama-3.1-8B; for Aya-23-8B the two roles are more balanced.
5. Interaction terms between source and target proxies are not statistically significant under any model, transform, or subset we tested. This means BLEU is well captured by main effects of per-language competence, with no detectable language-pair-specific component.

The data are pairwise BLEU scores and per-language proxy metrics for two open multilingual decoder models, Llama-3.1-8B and Aya-23-8B, over 24 languages from FLORES (552 directed pairs per model). Multi-BLiMP accuracy is available for 17 of those languages.

For the per-language correlation, direct BLEU prediction, source–target asymmetry, and interaction-term analyses (Sections 4–7), we restrict to the 17-language subset for which both proxies are available, so that all comparisons between FLORES PPL and Multi-BLiMP are on identical data.

Corresponding results on the broader 24-language FLORES set (and on the 23-language no-English variant) for FLORES PPL are reported in Appendix D.

We pay particular attention to a methodological subtlety throughout: English is a leverage point that distorts FLORES-PPL-based analyses for Llama, but not for Multi-BLiMP-based analyses. Main-body random-effects results therefore exclude English (Section 3); extended results with all languages and the English-outlier evidence are in Appendix A.

2 Setup

Models. Llama-3.1-8B (Meta) and Aya-23-8B (Cohere). Both are dense multilingual decoder LMs of comparable size.

BLEU. For each ordered language pair (i, j) with $i \neq j$, we use the sentencepiece-tokenized BLEU score of the model translating from language i into language j on FLORES devtest. This gives a 24×24 matrix per model with 24 undefined diagonal cells (552 valid cells per model).

Monolingual competence proxies. We consider two per-language scalars:

- FLORES corpus perplexity P_i : token-level PPL of the model on the FLORES devtest sub-corpus for language i .
- Multi-BLiMP accuracy a_i : proportion of minimal pairs in the Multi-BLiMP grammatical-acceptability dataset for language i that the model scores correctly. Available for 17 of 24 languages: `ces`, `deu`, `ell`, `eng`, `fas`, `fra`, `heb`, `hin`, `ita`, `nld`, `pol`, `por`, `ron`, `rus`, `spa`, `tur`, `ukr`.

R^2 on centered scale. Throughout we report standard regression $R^2 = 1 - \sum_{ij} (y_{ij} - \hat{y}_{ij})^2 / \sum_{ij} (y_{ij} - \bar{y})^2$, where $\bar{y}$ is the sample mean over the observed cells. The denominator is centered, so the grand mean is not credited as “explained”.

3 An additive random-effects model fits BLEU well

We fit the two-way additive random-effects model

$$\text{BLEU}_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}, \quad \sum_i \alpha_i = \sum_j \beta_j = 0, \quad (1)$$

to all observed (non-diagonal) cells of the BLEU matrix by OLS. The recovered α_i is a per-language source-side effect and β_j is a per-language target-side effect, both measured as deviations from the grand mean BLEU.

English is a leverage point. On the full 24-language set, English is a large positive outlier in both α and β (Appendix A). Its presence substantially distorts downstream PPL-based correlation analyses for Llama. We therefore report the random-effects fit on the 23-language subset that excludes English. Doing so does not change the headline conclusion — the additive fit is essentially as strong with English included as without — but it lets the per-language structure speak more cleanly across the remaining languages.

Fit quality. On the 23-language no-English subset ($n = 506$ observed pairs per model), the additive model explains 86–93% of centered BLEU variance with only $2(n_{\text{lang}} - 1) = 44$ free parameters per language axis (Table 1). With 1%–2% exception (Aya $\bar{y}$ shifts $10.81 \rightarrow 9.88$, Llama $13.27 \rightarrow 12.02$), the recovered μ and the additive R^2 are essentially indistinguishable from those on the full set.

model	$\bar{y}$	R^2_{additive}
Aya-23-8B	9.88	0.862
Llama-3.1-8B	12.02	0.927

Table 1: Additive random-effects model fit on the 23-language no-English subset, with $n = 506$ observed pairs per model. R^2 is on centered total sum of squares.

Multiplicative comparison. The natural alternative to an additive decomposition is multiplicative: $\text{BLEU}_{ij} \approx u_i v_j$, the rank-1 SVD approximation. To compare apples-to-apples we add a constant grand-mean offset to the multiplicative model so both have the mean given for free, and we parameter-match: additive uses $\mu + \alpha + \beta$ with sum-to-zero constraints; multiplicative uses $\bar{y} + u_i v_j$ with the rank-1 factors fit by SVD of the mean-centered matrix.

model	additive ($\mu + \alpha + \beta$)	multiplicative ($\bar{y} + uv$)	AMMI (additive + rank-1 of residuals)
Aya-23-8B	0.862	0.716	0.906
Llama-3.1-8B	0.927	0.866	0.956

Table 2: Centered R^2 for additive vs. multiplicative decompositions, and for AMMI (additive main effects plus rank-1 of the additive residuals), on the 23-language no-English subset. Additive and multiplicative use comparable parameter budgets; AMMI uses roughly twice as many.

Three conclusions follow from Table 2:

1. **Additive is the better-fitting class on this parameter budget.** Multiplicative fits the data well but does not outperform additive on either model. For Aya the gap is substantial (+14.6 pp); for Llama it is smaller (+6.1 pp) but still favors additive.
2. **Multiplicative is not negligible.** A rank-1 outer product captures 72–87% of centered variance; the per-language structure of BLEU is not orthogonal to a multiplicative parameterization.
3. **Multiplicative residual interaction is small.** AMMI — adding rank-1 of the additive residuals on top of the additive fit — improves R^2 by only 2.9 pp (Llama) and 4.4 pp (Aya). Beyond per-language main effects, genuinely multiplicative structure is a small contribution.

For the rest of the paper we use the α_i and β_j estimated from the additive model in equation (1) as per-language “competence” estimates: α_i is how much above (or below) the grand-mean BLEU language i scores as a translation source, and β_j analogously as a target.

4 Per-language effects align with monolingual competence proxies

Both FLORES corpus perplexity P_i and Multi-BLiMP accuracy a_i are per-language scalars meant to measure model competence in language i . If the additive language effects α, β recovered in

Section 3 reflect translation-relevant competence, we expect them to correlate with each proxy in the appropriate direction (negative with P , positive with a).

We compute these correlations on the 17-language Multi-BLiMP subset (which includes English, since English is not an outlier in the Multi-BLiMP framework — Appendix A). This subset gives the apples-to-apples comparison of the two proxies on the same languages. Corresponding results on the 23-language non-English subset for FLORES PPL are reported in Appendix D, and they are qualitatively consistent with what is reported here.

Table 3 summarizes the strongest correlation under each transform family for each model and role.

model	role	FLORES PPL ($ r $)			Multi-BLiMP accuracy ($ r $)	
		raw P	$\log P$	$1/P$	raw a	$\text{logit}(a)$
Aya-23-8B	src (α)	0.46	0.43	0.39	0.74	0.82
Aya-23-8B	tgt (β)	0.15	0.14	0.14	0.63	0.77
Llama-3.1-8B	src (α)	0.13	0.14	0.10	0.35	0.38
Llama-3.1-8B	tgt (β)	0.29	0.38	0.40	0.54	0.64

Table 3: Per-language correlations between additive language effects (α, β) and monolingual competence proxies, on the 17-language Multi-BLiMP subset. All values are absolute Pearson $|r|$; all best transforms in the indicated direction (negative for P and $\log P$, positive for $1/P$ and Multi-BLiMP transforms). Bold marks the strongest correlation in each row.

Three observations:

1. **Multi-BLiMP correlations are systematically stronger than FLORES PPL correlations.** For Aya, the best Multi-BLiMP correlation reaches $|r| = 0.82$ on α ($p < 0.001$ at $n = 17$); the best PPL correlation is $|r| = 0.46$. For Llama, Multi-BLiMP reaches $|r| = 0.64$ on β ($p < 0.01$); PPL reaches $|r| = 0.40$.
2. **The correlation patterns differ between the two models.** For Aya, α tracks both proxies more strongly than β does. For Llama, β tracks both proxies more strongly than α does. We return to this asymmetry in Section 6.
3. **Llama α is the weak point.** Even Multi-BLiMP achieves only $|r| \approx 0.38$ for Llama source effects, vs. $|r| \approx 0.64$ for Llama target effects. We interpret this as target-language generation being more directly downstream of the grammatical competence Multi-BLiMP measures, while source-language processing involves more content understanding.

Full tables across all four transforms tested for each proxy are in Appendix B, along with significance test results.

5 Monolingual competence proxies predict BLEU directly

We now ask whether monolingual proxies predict the pairwise BLEU score y_{ij} directly, fitting

$$y_{ij} = a_0 + b \cdot f(\text{proxy}_i) + c \cdot g(\text{proxy}_j) + \varepsilon_{ij} \quad (2)$$

by OLS on all valid pairs in the 17-language subset (272 pairs for Llama, 240 for Aya after dropping pairs missing either proxy). For PPL we test the raw- P form and a log-log analog of equation (2); for Multi-BLiMP we test raw- a and the log-error-rate analog $\log(1 - a)$.

Two takeaways from Table 4:

specification	Aya R^2	Llama R^2
$\text{BLEU} \sim P_s + P_t$	0.061	0.071
$\log \text{BLEU} \sim \log P_s + \log P_t$	0.068	0.127
$\text{BLEU} \sim a_s + a_t$	0.394	0.250
$\log \text{BLEU} \sim \log(1 - a_s) + \log(1 - a_t)$	0.444	0.339

Table 4: Direct prediction of (log-)BLEU from per-language competence proxies, on the 17-language Multi-BLiMP subset. Both PPL coefficients are highly significant ($p < 10^{-4}$) for Aya in both PPL forms; for Llama the $\log P_s$ coefficient is not significant ($p = 0.38$ in the log specification), while $\log P_t$ is highly significant ($p < 10^{-9}$). Both Multi-BLiMP coefficients are highly significant ($p < 10^{-2}$) in every cell.

1. **Multi-BLiMP is a strong predictor of BLEU.** The best Multi-BLiMP-based specification explains 44% (Aya) and 34% (Llama) of pairwise BLEU variance using just two per-language scalars and no interaction. These are large effects for a model with only two free coefficients beyond an intercept.
2. **FLORES PPL is a moderate predictor.** The best PPL specification reaches only $R^2 = 0.07$ (Aya) and $R^2 = 0.13$ (Llama) on the same subset. Multi-BLiMP yields 3–6× the explanatory power of PPL on the same data.

The gap between PPL and Multi-BLiMP is consistent with the per-language correlation results of Section 4: the same per-language structure that drives Multi-BLiMP’s stronger $|r|$ with α, β shows up here as stronger pairwise predictive power.

Coefficient signs are in the expected direction throughout: higher PPL or lower Multi-BLiMP accuracy on either side of the translation pair predicts lower BLEU. We report per-coefficient values and standard errors in Appendix E.

6 Source vs. target competence

The per-language effects α (source) and β (target) are both informative about BLEU, but they are not equally informative, and their relative roles differ between the two models.

Variance. Across pairs, the variance of β over the BLEU matrix is substantially larger than the variance of α for both models: roughly 4× for Aya and 9× for Llama. Choosing a different target language moves BLEU substantially more than choosing a different source language. This means that, in any per-pair regression of BLEU on proxies, the target side has more variance to absorb and tends to dominate the fit.

Correlation with competence proxies. On the 17-language Multi-BLiMP subset (Table 3):

- **Llama is target-dominant.** β correlates with Multi-BLiMP at $|r| = 0.64$ (logit transform), vs. $|r| = 0.38$ for α . With FLORES PPL, the pattern is the same in direction: $|r| = 0.40$ for β vs. $|r| = 0.13$ for α .
- **Aya is more symmetric.** Multi-BLiMP correlations are comparable on the two sides: $|r| = 0.82$ for α vs. $|r| = 0.77$ for β . With PPL the asymmetry shifts the other way ($|r| = 0.46$ for α vs. $|r| = 0.15$ for β), but both sides remain in the same range under Multi-BLiMP.

Interpretation. Translation as a target task is dominated by generating grammatically-well-formed output in the target language; this is exactly the kind of grammatical competence that Multi-BLiMP measures. Translation as a source task additionally requires content understanding, which a per-language grammaticality test does not measure. The fact that Llama shows a stronger target/source gap than Aya is consistent with the two models having different multilingual training distributions; we do not attempt to attribute the difference further here.

7 Interaction terms are not significant

A multiplicative interaction term in equation (2) — the simplest way to introduce language-pair-specific structure into the regression — would be evidence against the strict “main effects only” reading of H1. We test for it directly.

Specifically, on the 17-language Multi-BLiMP subset (and separately on its no-English subset), we fit the saturated model

$$y_{ij} = a_0 + b \cdot f(\text{proxy}_i) + c \cdot g(\text{proxy}_j) + d \cdot f(\text{proxy}_i) \cdot g(\text{proxy}_j) + \varepsilon_{ij} \quad (3)$$

and compare it to the main-effects-only version (equation (2)) via a nested-model F -test for the coefficient d .

subset	predictor (model)	$F(1, \text{df})$	p -value	ΔR^2
17-lang (Multi-BLiMP)	Multi-BLiMP, Aya	0.00	0.99	$< 10^{-4}$
17-lang (Multi-BLiMP)	Multi-BLiMP, Llama	0.56	0.46	0.001
17-lang (Multi-BLiMP)	PPL, Aya	0.09	0.76	$< 10^{-3}$
17-lang (Multi-BLiMP)	PPL, Llama	0.01	0.92	$< 10^{-3}$
16-lang (no English)	Multi-BLiMP, Aya	0.11	0.75	0.0003
16-lang (no English)	Multi-BLiMP, Llama	0.43	0.51	0.001
16-lang (no English)	PPL, Aya	0.001	0.97	$< 10^{-4}$
16-lang (no English)	PPL, Llama	0.16	0.69	0.001

Table 5: Nested-model F -test for the source–target interaction coefficient d in equation (3), on the 17-language Multi-BLiMP subset (top) and the same subset with English removed (bottom). The interaction is uniformly non-significant.

The interaction F -test fails to reject the null in every cell of Table 5: the smallest p -value is 0.46, and the increment ΔR^2 from adding the interaction term is at most 0.001. Adding interaction terms in the raw vs. log form, on the full 24-language vs. the 17-language subset, with or without English, with PPL or Multi-BLiMP — none of these surfaces detectable pair-specific structure beyond what main effects already capture.

This is direct evidence for H1: pairwise BLEU is well predicted by *main effects* of per-language competence, and the additional predictive power from a pair-specific interaction is below the noise threshold of our data.

8 Discussion

The evidence reviewed here supports a strong form of H1: pairwise BLEU scores between language pairs are well captured by an additive decomposition into per-language source and target effects,

those effects align with monolingual competence proxies in the expected direction, and a per-language proxy alone predicts pairwise BLEU up to $R^2 = 0.44$. Adding language-pair-specific interaction terms does not improve the fit. To the extent there is any residual structure beyond per-language main effects, it is small (a few percentage points of R^2 , captured by AMMI’s rank-1 residual).

The most surprising and arguably most actionable finding is the gap between FLORES corpus PPL and Multi-BLiMP as competence proxies. Multi-BLiMP achieves $3\text{--}6\times$ the predictive R^2 of PPL on the same pairs. This suggests that, for the purpose of predicting translation quality, a targeted measure of grammatical competence is substantially more informative than the more commonly reported corpus-level PPL.

A English as an outlier in PPL-based analyses

This appendix documents the leverage effect of English on FLORES-PPL-based correlations, and shows that English is *not* a leverage point in Multi-BLiMP-based analyses.

English in the additive fit. After fitting equation (1) on the full 24-language set, the recovered α and β for English are large positive outliers: $\alpha_{\text{eng}} = +8.47$ and $\beta_{\text{eng}} = +21.07$ for Llama; $\alpha_{\text{eng}} = +7.62$ and $\beta_{\text{eng}} = +13.46$ for Aya (signed residuals to the linear fit of each effect on $\log P$). These are substantially larger in magnitude than the next-largest residuals, which range from 4 to 12 in absolute value.

Effect on per-language PPL correlations. Table 6 shows the Pearson correlation between α, β and PPL transforms with and without English in the sample.

model	transform	Pearson r on α (src)			Pearson r on β (tgt)		
		full ($n = 24$)	no eng ($n = 23$)	$\Delta r $	full ($n = 24$)	no eng ($n = 23$)	$\Delta r $
Llama	raw P	−0.19	− 0.44 *	+0.25	−0.15	−0.34	+0.20
Llama	log P	−0.14	−0.36	+0.22	−0.04	−0.19	+0.15
Llama	$1/P$	+0.10	+0.26	+0.16	−0.08	+0.01	−0.07
Aya	raw P	−0.30	−0.31	+0.01	−0.36	−0.39	+0.03
Aya	log P	−0.36	−0.40	+0.04	−0.34	−0.39	+0.05
Aya	$1/P$	+0.40	+ 0.48 *	+0.08	+0.32	+0.38	+0.06

Table 6: Pearson correlations between α, β and PPL transforms on the full 24-language set vs. the 23-language no-English set. $\Delta|r| = |r_{\text{no eng}}| - |r_{\text{full}}|$. * $p < 0.05$.

For Llama, removing English roughly doubles the magnitude of Pearson correlations across all PPL transforms and brings $r(\alpha, P)$ to statistical significance ($p = 0.035$). For Aya the effect is small (+1 to +8 pp): the Aya signal was already systematic across all 24 languages.

Direct BLEU regression. The same outlier effect appears in the direct PPL-to-BLEU regression on the full 24-language pair set: removing English moves Llama from $R^2 = 0.021$ (full set) to $R^2 = 0.114$ (no English), a $5.5\times$ improvement, with both PPL coefficients becoming highly significant. Aya improves modestly: $R^2 = 0.105 \rightarrow 0.120$.

English is not an outlier for Multi-BLiMP. By contrast, the analogous correlations using Multi-BLiMP accuracy are essentially unchanged when English is removed from the 17-language subset: Aya $|r(\alpha, \text{logit } a)|$ shifts $0.82 \rightarrow 0.77$; Llama $|r(\beta, \text{logit } a)|$ shifts $0.64 \rightarrow 0.49$. Removing English even slightly weakens some Llama Multi-BLiMP correlations, because English is a valid high-accuracy, high-BLEU point that anchors the relationship. Multi-BLiMP correctly orders English with its high BLEU; FLORES corpus PPL does not.

This contrast is the methodological justification for two choices in the main body: (i) random-effects results in Section 3 exclude English, so that the per-language effects are reported on a sample where PPL-based downstream analyses will not be artificially weakened; (ii) downstream analyses in Sections 4–7 use the 17-language Multi-BLiMP subset (which includes English), since in that framework English is not an outlier.

B Full per-language correlation tables (17-language subset)

model	role	transform	Pearson r	Pearson p	Spearman ρ	n
Multi-BLiMP accuracy (a)						
Aya	src (α)	raw	+0.74	0.001	+0.72	17
Aya	src (α)	$1 - a$	−0.74	0.001	−0.72	17
Aya	src (α)	$\log(1 - a)$	−0.82	0.0001	−0.72	17
Aya	src (α)	$\text{logit}(a)$	+0.82	0.0001	+0.72	17
Aya	tgt (β)	raw	+0.63	0.008	+0.67	17
Aya	tgt (β)	$\log(1 - a)$	−0.78	0.0004	−0.67	17
Aya	tgt (β)	$\text{logit}(a)$	+0.77	0.0005	+0.67	17
Llama	src (α)	raw	+0.35	0.16	+0.31	17
Llama	src (α)	$\log(1 - a)$	−0.38	0.13	−0.31	17
Llama	src (α)	$\text{logit}(a)$	+0.38	0.13	+0.31	17
Llama	tgt (β)	raw	+0.54	0.024	+0.60	17
Llama	tgt (β)	$\log(1 - a)$	−0.65	0.005	−0.60	17
Llama	tgt (β)	$\text{logit}(a)$	+0.64	0.005	+0.60	17
FLORES PPL (P)						
Aya	src (α)	raw P	−0.46	0.064	—	17
Aya	src (α)	$\log P$	−0.43	0.087	—	17
Aya	src (α)	$1/P$	+0.39	0.117	—	17
Aya	tgt (β)	raw P	−0.15	0.57	—	17
Aya	tgt (β)	$\log P$	−0.14	0.58	—	17
Aya	tgt (β)	$1/P$	+0.14	0.58	—	17
Llama	src (α)	raw P	+0.13	0.63	—	17
Llama	src (α)	$\log P$	+0.14	0.60	—	17
Llama	src (α)	$1/P$	−0.10	0.71	—	17
Llama	tgt (β)	raw P	+0.29	0.25	—	17
Llama	tgt (β)	$\log P$	+0.38	0.13	—	17
Llama	tgt (β)	$1/P$	−0.40	0.11	—	17

Table 7: Per-language correlations on the 17-language Multi-BLiMP subset. For Multi-BLiMP transforms we list all four tested. For PPL the four transforms tested are raw, log, $1/P$, and $1/\sqrt{P}$; the latter is omitted for compactness (results are nearly identical to $1/P$). Note that several Llama α -vs-PPL correlations have the “wrong” sign on this subset; this is the leverage effect of English documented in Appendix A expressed on a different subset.

C Model specifications and inferential procedures

C.1 Additive random-effects model

Fit by ordinary least squares on the long-format pair data, with dummy coding for source and target language identifiers. To recover sum-to-zero α, β from a non-singular design we drop one source dummy and one target dummy as references, then re-center the recovered coefficients post-hoc by subtracting the mean of each α and β vector and absorbing those means into μ .

Parameter count on n_{lang} languages with sum-to-zero constraints: $1 + (n_{\text{lang}} - 1) + (n_{\text{lang}} - 1) = 2n_{\text{lang}} - 1$. For $n_{\text{lang}} = 23$ this is 45; for $n_{\text{lang}} = 17$ this is 33. The residual degrees of freedom for the 23-language fit are $506 - 45 = 461$; for the 17-language fit, $272 - 33 = 239$ (Llama) or $240 - 33 = 207$ (Aya, after dropping pairs with missing PPL).

C.2 Multiplicative (rank-1) decomposition

The matrix is mean-centered to $M_c = M - \bar{y}$ on its observed cells; missing diagonal cells are imputed with column means prior to SVD. The rank-1 SVD $u_1 v_1^\top$ of M_c gives the multiplicative factors. The model prediction for cell (i, j) is $\bar{y} + (u_1)_i (v_1)_j$. Parameter count (with the scale absorbed once) is $2n_{\text{lang}} - 1$, matching the additive model. We evaluate R^2 on the observed cells only.

C.3 AMMI

After fitting the additive model, we compute the residual matrix $M - (\mu + \alpha + \beta)$ on all observed cells (with the same diagonal imputation as above for the SVD step), take its rank-1 SVD, and add this back to the additive prediction. Total parameter count is roughly twice the additive model’s.

C.4 F -test for the source–target interaction term

For a nested-model comparison with one additional parameter:

$$F = \frac{(\text{RSS}_{\text{add}} - \text{RSS}_{\text{int}})/1}{\text{RSS}_{\text{int}}/(n - p_{\text{int}})} \sim F(1, n - p_{\text{int}})$$

where p_{int} is the number of free parameters in the interaction model and n is the number of observed pairs. The F -test is equivalent to a two-sided t -test on the interaction coefficient when only one parameter is added.

C.5 Per-coefficient inference

For coefficient $\hat{\beta}_k$ from an OLS fit with residual variance $\hat{\sigma}^2 = \text{RSS}/(n - p)$ and design matrix X , the standard error is $\widehat{SE}(\hat{\beta}_k) = \hat{\sigma} \sqrt{(X^\top X)^{-1}_{kk}}$ and the t -statistic is $\hat{\beta}_k / \widehat{SE}(\hat{\beta}_k)$. Two-sided p -values are computed against a t -distribution with $n - p$ degrees of freedom.

D FLORES PPL on the 23-language non-English subset

For completeness, we report the PPL-only versions of Sections 4 and 5 on the 23-language non-English subset (i.e., the 24 FLORES languages minus English, not restricted to the Multi-BLiMP set). This is the subset referenced as “the larger PPL sample” in the main body.

model	transform	$ r $ on α	$ r $ on β
Aya	raw P	0.31	0.39
Aya	$\log P$	0.40	0.39
Aya	$1/P$	0.48*	0.38
Llama	raw P	0.44*	0.34
Llama	$\log P$	0.36	0.19
Llama	$1/P$	0.26	0.01

Table 8: Per-language Pearson $|r|$ on the 23-language no-English subset. * $p < 0.05$.

Per-language correlations. On 23 non-English languages (Table 8), Pearson correlations between α, β and PPL transforms are $|r| \approx 0.34$ – 0.45 for both models, with several reaching $p < 0.05$.

Direct BLEU prediction. On the 23-language no-English pair set (506 pairs per model), the direct PPL→BLEU regressions reach $R^2_{\text{Llama}} = 0.114$ (raw form) and $R^2_{\text{Aya}} = 0.145$ (log form), with both PPL coefficients highly significant in all four (model, form) cells. These are similar in magnitude to the 17-language subset numbers in Section 5, confirming that the 17-language restriction is not the source of the PPL→BLEU R^2 being moderate rather than large.

E Per-coefficient details for the BLEU $\sim$ proxy fits

model	specification	$\hat{b}$ (src)	p (src)	$\hat{c}$ (tgt)	p (tgt)
Aya	$\text{BLEU} \sim P_s + P_t$	−0.13	$5.8 \cdot 10^{-4}$	−0.06	0.046
Aya	$\log \text{BLEU} \sim \log P_s + \log P_t$	−0.61	$6.9 \cdot 10^{-5}$	−0.16	0.244
Aya	$\text{BLEU} \sim a_s + a_t$	+50.6	$2.1 \cdot 10^{-11}$	+75.9	$5.2 \cdot 10^{-22}$
Aya	$\log \text{BLEU} \sim \log(1 - a_s) + \log(1 - a_t)$	−1.21	$3.3 \cdot 10^{-14}$	−1.93	$4.4 \cdot 10^{-25}$
Llama	$\text{BLEU} \sim P_s + P_t$	−0.05	0.39	−0.42	$1.1 \cdot 10^{-5}$
Llama	$\log \text{BLEU} \sim \log P_s + \log P_t$	−0.10	0.37	−1.05	$1.6 \cdot 10^{-9}$
Llama	$\text{BLEU} \sim a_s + a_t$	+18.6	$6.8 \cdot 10^{-3}$	+90.4	$9.4 \cdot 10^{-18}$
Llama	$\log \text{BLEU} \sim \log(1 - a_s) + \log(1 - a_t)$	−0.45	$4.2 \cdot 10^{-3}$	−2.39	$2.9 \cdot 10^{-25}$

Table 9: Per-coefficient estimates, with two-sided OLS p -values, on the 17-language Multi-BLiMP subset. Sample size: 272 (Llama), 240 (Aya), after dropping pairs missing either proxy on either side. All signs are in the expected direction (negative for P and $\log(1 - a)$; positive for a).